How to adjust the setting for multiagent simulations

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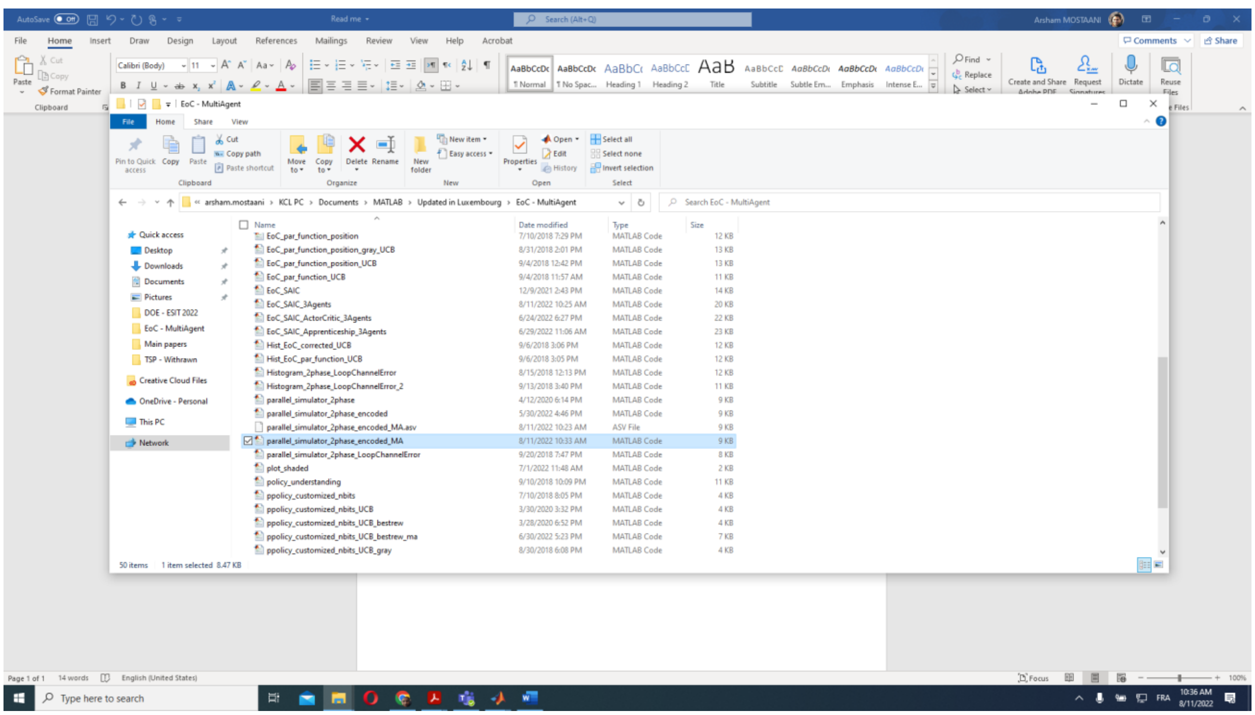
ESAIC – SAIC

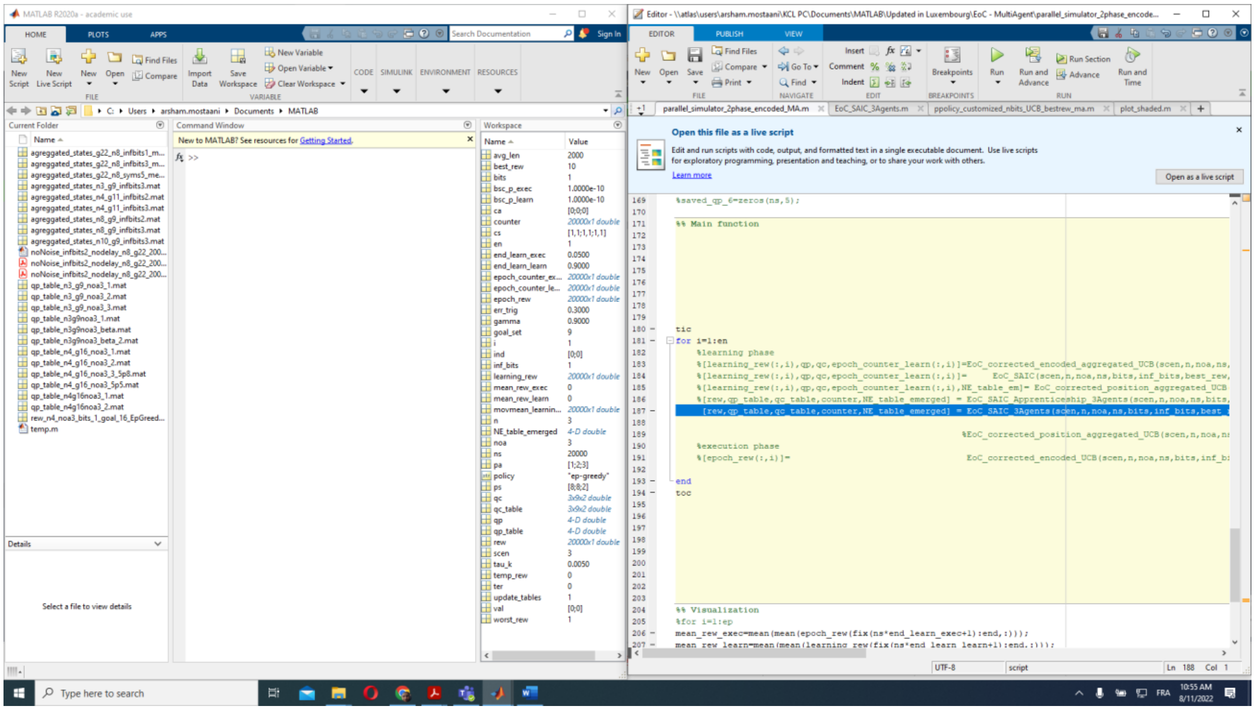
Finding the right functions and scripts

1- The controller script is "Parallel_simulator_2phase_encoded_MA"



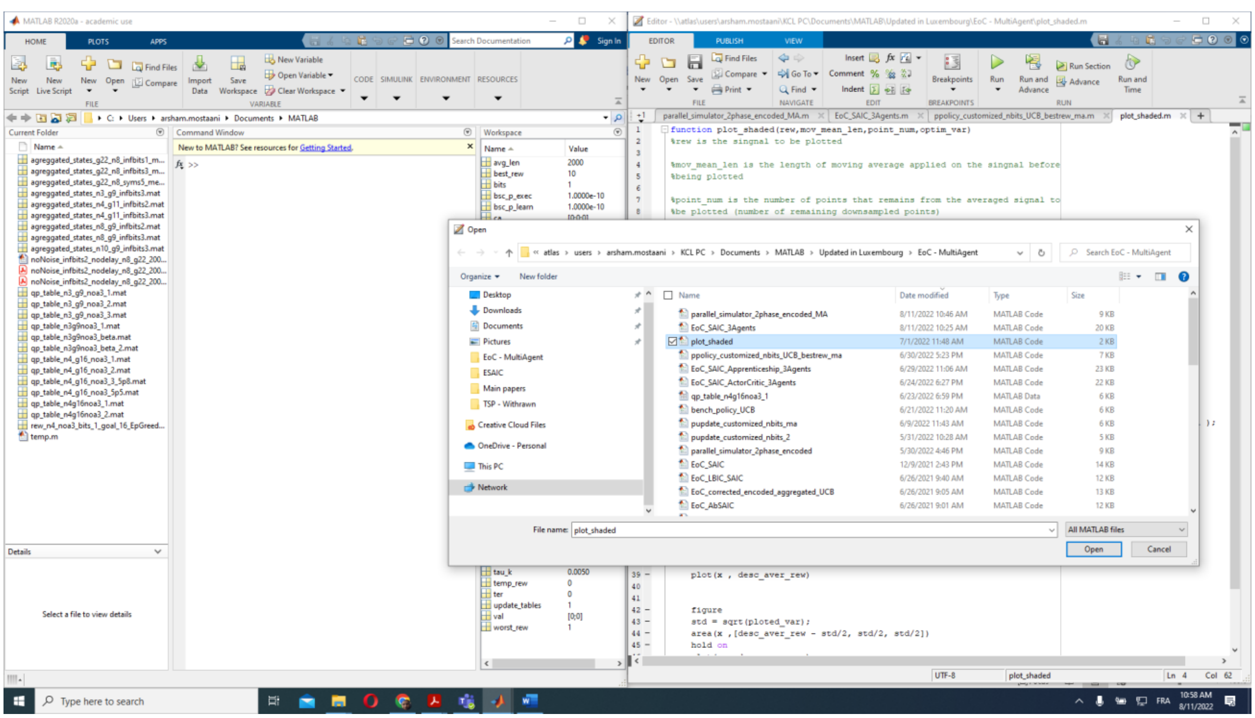
- The controller script is control panel that gives the user all sorts of control facilities to adjust the simulations to their needs.

2- The simulator wrapper is " EoC_SAIC_3Agents "



- The simulator wrapper is wrapper is a function that carries out the simulation for one complete epoch

3- To have nice looking simulation results you can also use the “plot_shaded” function, which plots the reward signal together with a shaded area of its variance.



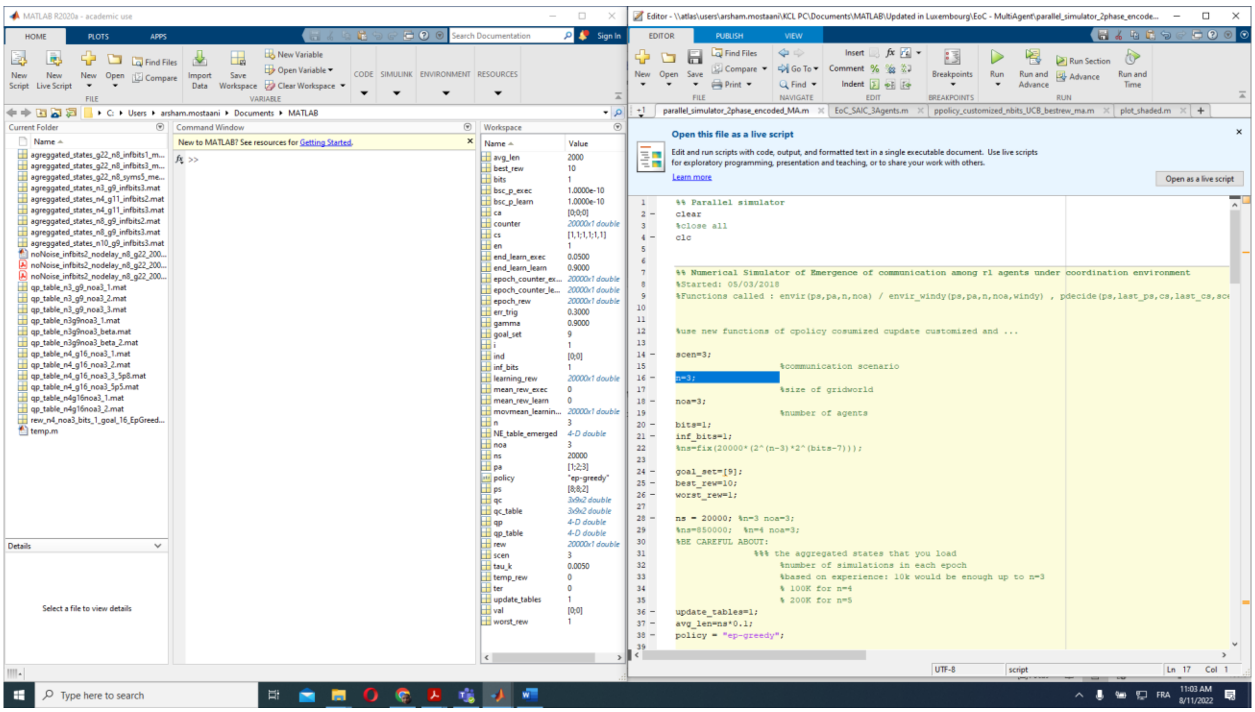
4- All these scripts and functions are available in the folder:

\\atlas\users\arsham.mostaani\KCL PC\Documents\MATLAB\Updated in Luxembourg\EoC - MultiAgent

This folder is on the server of the university and is frequently backed up.

Adjustment of the control panel

1- Select the number of grids in each direction e.g., $n = 3$ is selected when you want a 3x3 grid world



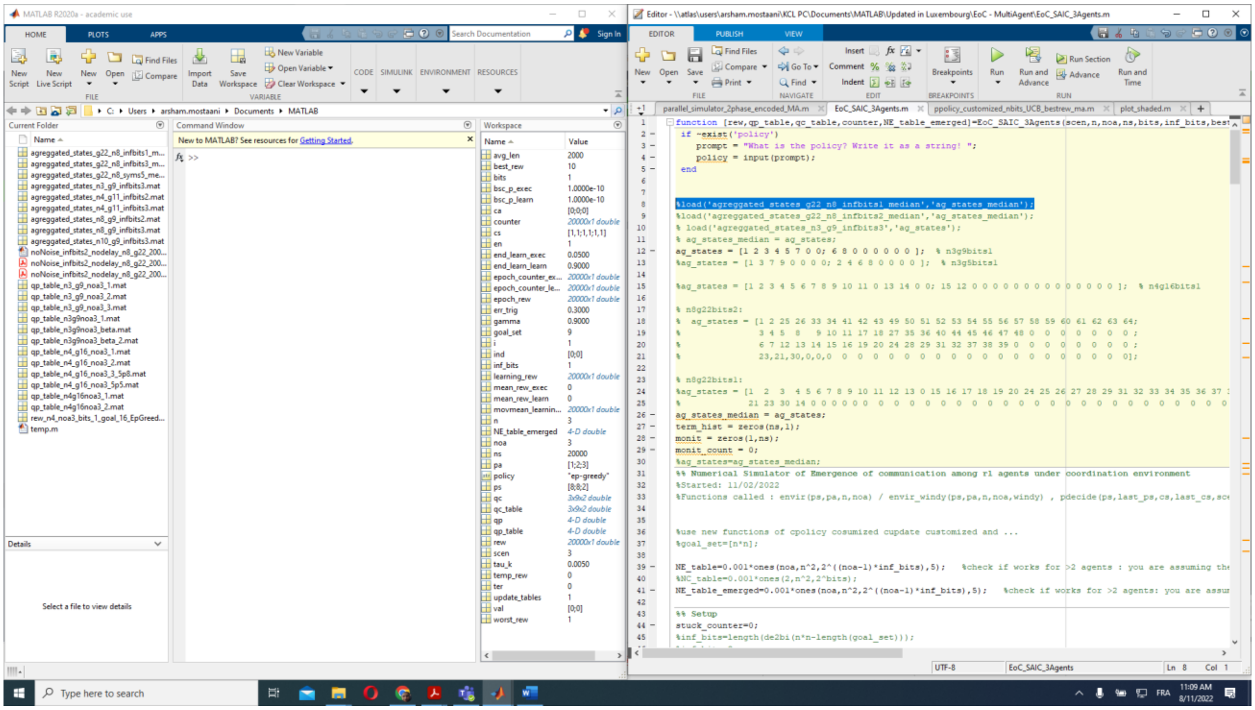
2- Select the number of agents with the parameter noa (standing for the Number of Agents)

```
noa=3;
```

3- Select the number of communication bits – in the case of task-oriented quantization bits and inf_bits should have similar values

```
bits=1;
inf_bits=1;
```

4- According to the size of the grid-world and the number of quantized bits, load the right quantizer inside the wrapper



5- Select the right goal set/location

```
goal_set=[9];
```

6- Update tables = 1 – This parameter should only be zero when using Apprenticeship learning wrapper.

```
update_tables=1;
```

7- Choose the right policy – we have obtained very good results with decaying epsilon greedy:

```
policy = "ep-greedy"
```

8- Check your rewarding scheme – the default setting is adjusted according to our paper: "Task-Oriented Data Compression for Multi-Agent Communications Over Bit-Budgeted Channels" – This is the TSP paper being withdrawn and submitted to OJCOMS. Accordingly, the common reward signal increases exponentially as the number of agents arrived at the goal point are increased.